

CÓDIGO:

Question 6. A cumbersome alloy... (10% of the total; only 10th grade)

6.1	6.2	Total	Total
10	30	40	10%

Stoichiometry emerged in the 18th century when the basic principles governing chemical processes were stated, including the law of conservation of mass, and laid the foundations of modern chemistry.

In the following problem, you must apply your knowledge of algebra and stoichiometry.

A 10.0 g sample of an alloy contains three metals of the first transition series (nickel, X and Y) in its composition. The sample is reacted with sufficient HCl, obtaining a solution that contains the chlorides of the metals Ni and X, with formulas NiCl_2 and XCl_2 , in addition to 0.0529 mol of H_2 . And it does not react with hydrochloric acid. NaOH is subsequently added to the solution, forming the reddish-brown hydroxide. On the other hand, when the remaining solid Y is heated in air, 8.701 g of the corresponding oxide of formula YO is formed.

6.1. Set up the **balanced** equations for the chemical reactions mentioned previously.

6.2. Identify metals **X** and **Y** and calculate the alloy composition (mass %).

Question 7. Camagüey es la ciudad... (10% of the total; only 11th grade)

7.1	7.2	7.3	Total	Total
2	12	26	40	10%

The “tinajón” is the Camagüey symbol par excellence; it is the most rooted local representation, which is why Camagüey is known as the City of the Tinajones. The chemical composition

of a tinajón is interesting. Usually, the material contains three fundamental components and 2% impurities. When the Ibero-American contestants Lianet and Daniela began chemistry in their province in tenth grade, they became interested in the composition of a jar. They discovered that it contains an atomic solid with a covalent bond that is abundant in the Earth’s crust.

7.1. Could you tell what solid it is?

Daniela then inquired about kaolin, one of the significant components of the tinajones. Kaolin is a mixed oxide that has stoichiometric amounts (equal amount of substance) of aluminum and silicon, in addition to 55.78% (m/m) of oxygen and 1.56% (m/m) of hydrogen.

7.2. Calculate the empirical formula of kaolin, considering that its formula mass is 258.26 g/mol.

For her part, Lianet was fascinated with micas, which can appear in different proportions in the same jar. The generic composition of some micas can be expressed by the generic formula $\text{X}_2\text{Y}_{4-6}\text{Z}_8\text{O}_{20}(\text{OH})_4$,

where **X** is K or Na, **Y** is Al, Mg or Fe and **Z** is Si or Al, fundamentally. In a workshop near Lianet’s house, three micas are sold. Their elemental analysis is shown below.

Mica	Components (% in mass)
I	O (47,27); Si (27,67); Mg (14,95); K (9,62); H (0,49)
II	O (39,00); Fe (34,00); Al (21,93); OH (6,90); Na (4,67)
III	O (48,39); Al(6,81); Si (28,33); K (9,85); Mg (6,12); H (0,50)

7.3. Calculate the formulas of micas I, II and III. **Please note the generic composition given above.**

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Question 8. CSI in the chemical laboratory (10% of the total; only 10th grade)

8.1	8.2	8.3	Total	Total
10	20	10	40	10%

Luis Manuel and Adriana are two students who were part of the 2023 Chemistry Pre-selection. On one occasion, they detected that the labels of five reagent bottles were damaged.

Not being able to read them and identify the substances in these bottles, they decided to consult the list of entry and exit of reagents from the warehouse. With this document, they determined that the substances were **barium carbonate**, **calcium sulfate**, **lead(II) acetate**, **potassium cyanide** and **sodium sulfide**. They immediately set about investigating what substance was in each bottle. To do this, they labeled each one with a different alphabet letter.

- Solids **A**, **C**, **D** and **E** were white, while solid **B** was pale yellow.
- **A**, **C** and **E** were odorless, while **B** smelled like rotten eggs and **D** smelled like bitter almonds.
- Compounds **A** and **C** were insoluble in water, while the others dissolved well.

At this point, the students were almost sure what substances **B**, **D**, and **E** were.

- Solid portions of **B**, **C** and **D** were dissolved in aqueous hydrochloric acid, forming a gas. In contrast, **A** was insoluble, **E** was only slightly soluble, and neither evolved a gas.

Then, the two students could identify each substance, although they performed additional tests.

- The addition of sulfuric acid to the aqueous solution of **E** formed a white precipitate. In comparison, the addition of sodium hydroxide formed a white precipitate, soluble in excess reagent.
- The mixture of aqueous solutions of **B** and **E** made it possible to obtain a black solid.
- A precipitate with a characteristic dark blue color was obtained by adding **D** to an aqueous solution of iron(II) and iron(III) chlorides.

8.1. Identify the solids in jars **A**, **B**, **C**, **D**, and **E** by their chemical formula.

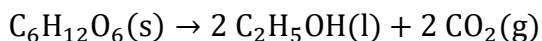
8.2. Formulate the adjusted equations of the chemical reactions carried out during the experiments.

8.3. In reality, the technicians could have identified each of the substances simply through the “flame test.” What colors characterize the cations of the previous salts?

Question 9. Chemicals get along well with ethanol (12% of the total; only 10th grade)

9.1	9.2	9.3	9.4	9.5	9.6	9.7	9.8	9.9	9.10	Total	Total
4	2	5	6	3	8	6	4	2	8	48	12%

Although biotechnology today is associated with the manufacture of vaccines and medicines, the use of living organisms to obtain compounds of interest is part of the early history of human civilization. One of the first compounds obtained was ethanol, used for recreational or medical purposes. Ethanol ($\text{CH}_3\text{-CH}_2\text{-OH}$, also formulated as $\text{C}_2\text{H}_5\text{OH}$) can be obtained from the fermentation of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) by different microorganisms:



Two students who participated in the last Central American and Caribbean Chemistry Olympiad, Diego and Royman, decided to synthesize ethanol. To do this, they took a 10.0 L container, filled it halfway with a glucose solution and placed it to ferment in a dark place at room temperature.

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Because they did not have the appropriate devices to allow the extraction of the gas formed during the reaction, one of their concerns was that the container had the proper strength for the process to be carried out.

9.1. Knowing that the container can withstand a pressure of 12 bar, calculate the maximum mass of ethanol ($M = 46.1 \text{ g/mol}$) that the students will produce at 25°C .

9.2. The ethanol obtained in this way is accompanied by various impurities. What separation method should students use to purify it? What physical property is it based on?

Global ethanol production has increased because it is a renewable fuel to produce energy, and the depletion of fossil fuels such as coal (C), oil and natural gas, mainly composed of methane (CH_4).

9.3. State the equations for the complete combustion of 1 mole of ethanol, methane and carbon with dioxygen, forming carbon dioxide gas and liquid water.

9.4. Represent the structures of ethanol, methane, dioxygen, carbon dioxide, and water.

Ethanol is miscible with water, meaning it can be mixed in any proportion and form a solution. On the contrary, dioxygen is slightly soluble in water, and carbon dioxide is somewhat more soluble than water.

9.5. Justify the different solubility in water of the substances ethanol, dioxygen and carbon dioxide based on the structures given in **9.4.** and intermolecular interactions.

9.6. From the values of the average bond energies and the other thermodynamic quantities, estimate the values of the enthalpy change of the previous reactions in **9.3.**

Bond	O=O	C=O	C–O	C–C	C–H	O–H
E(A-B, kJ/mol)	495	799	355	348	414	463

$$\Delta H_{\text{atom}}^\circ(\text{C}) = 717 \text{ kJ/mol}, \Delta H_{\text{vap}}^\circ(\text{ethanol}) = 42,4 \text{ kJ/mol}, \Delta H_{\text{vap}}^\circ(\text{H}_2\text{O}) = 40,7 \text{ kJ/mol}$$

$$\Delta_r H^\circ \approx \sum E(\text{broken bonds}) - \sum E(\text{formed bonds}) \quad [\text{For reactions in gas phase}]$$

You must calculate $\Delta H_{\text{comb}}^\circ(\text{ethanol})$, $\Delta H_{\text{comb}}^\circ(\text{CH}_4)$ and $\Delta H_{\text{comb}}^\circ(\text{C})$ in kJ/mol.

9.7. Calculate the previous values again (thermodynamically exact) using the enthalpies of formation corresponding to reactants and products.

$$\Delta H_f^\circ (\text{kJ/mol}): \text{ethanol} = -275; \text{CH}_4 = 79; \text{H}_2\text{O} = -286; \text{CO}_2 = -394.$$

You must calculate $\Delta H_{\text{comb}}^\circ(\text{ethanol})$, $\Delta H_{\text{comb}}^\circ(\text{CH}_4)$ and $\Delta H_{\text{comb}}^\circ(\text{C})$ in kJ/mol.

9.8. Calculate the gravimetric (MJ/kg) and volumetric (MJ/L) energy densities of ethanol. Consider that $\Delta H_{\text{comb}}^\circ(\text{ethanol}) = -1400 \text{ kJ/mol}$ if you could not calculate it before and density 0.789 g/mL .

9.9. Given the gravimetric and volumetric energy densities of charcoal (26-33 MJ/kg, 34-43 MJ/L) and methane (55.6 MJ/kg, 0.0378 MJ/L; at 101.3 kPa and 15°C). Consider your calculated values for ethanol and decide which of the three fuels is most efficient.

9.10. Royman and Diego decided to produce enough ethanol to heat all the water used in the first pre-selection stage for personal hygiene. Suppose that it is necessary to heat the water from 25°C

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to 40 °C, that there are 150 people in the training center for 45 days and that each person consumes an average of 30 L of water. How much ethanol do they need? The heat capacity of water is 4.186 kJ/(°C·kg) and its density is 1.00 g/mL. Consider $\Delta H_{\text{comb}}^\circ(\text{C}_2\text{H}_6\text{O}) = -1400 \text{ kJ/mol}$ if you were unable to calculate it above.

Question 10. Kinetics applied to enzymatic reactions (8% of the total; only 10th grade)

10.1	10.2	10.3	10.4	10.5	Total	Total
4	8	2	6	12	32	8%

Enzymes are the rapid “mediators” of chemical reactions in our bodies. Enzymes bind to molecules (substrates) at specific

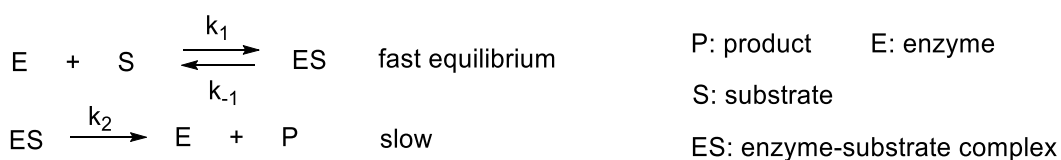
locations called catalytic sites and enable reactions to occur quickly. Studying these processes allows us to understand how our body metabolizes different substrates, which helps design disease treatments!

10.1. For a particular enzymatic reaction, it is known that its speed doubles with respect to the value of the speed at room temperature when the temperature increases by 10 °C. Considering that the system has Arrhenius-type behavior, calculate the activation energy (in kJ/mol).

10.2. Hydrogen peroxide is a very toxic metabolic product. The enzyme catalase catalyzes the breakdown of hydrogen peroxide into harmless water and dioxygen. Find the rate law and the value of the experimental specific rate constant from the initial rates and concentrations data in the table. **E** symbolizes the enzyme catalase.

v_0 (μMs^{-1})	$[\text{H}_2\text{O}_2]_0$ (μM)	$[\text{E}]_0$ (μM)
$2,0 \cdot 10^7$	10,0	0,50
$8,0 \cdot 10^6$	5,0	0,20
$2,0 \cdot 10^7$	8,5	0,50
$1,2 \cdot 10^7$	10,0	0,30

A mathematical model called “Michaelis-Menten kinetics” describes the kinetic behavior of enzymatic reactions. In 1913, Michaelis and Menten assumed an equilibrium between the enzyme, the substrate and the enzyme-substrate complex, formed instantaneously and maintained over time. The dissociation of the enzyme-substrate complex into enzyme and product is too slow to alter this balance. The model is summarized below:



The relationship $K_M = \frac{k_{-1} + k_2}{k_1}$ is called the Michaelis constant and in some cases, it numerically represents the inverse of the affinity of the substrate for the enzyme.

1. Assuming rapid chemical equilibrium: $k_1[\text{E}][\text{S}] = k_{-1}[\text{ES}]$

2. Separating constants from variables:

$$\frac{[\text{E}][\text{S}]}{[\text{ES}]} = \frac{k_{-1}}{k_1} = K_S \quad \text{where } K_S \text{ is the dissociation constant of the ES complex}$$

3. Substituting $[\text{E}] = [\text{E}]_0 - [\text{ES}]$, What is the mass balance equation and where $[\text{E}]_0$ is the initial (total) concentration of the enzyme

$$\frac{([\text{E}]_0 - [\text{ES}])[\text{S}]}{[\text{ES}]} = K_S$$

4. Solving for the concentration of the $[\text{ES}]$ complex, which is very difficult to measure experimentally:

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$$[ES] = \frac{[E]_0[S]}{K_S + [S]}$$

5. Since the dissociation step is the slow one, we have $v = k_2[ES]$ and therefore:

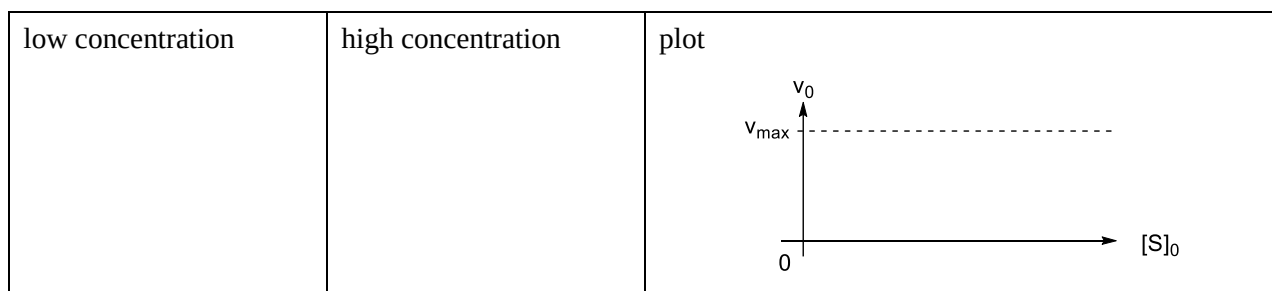
$$v = \frac{k_2[E]_0[S]}{K_S + [S]} \quad (1)$$

10.3. Why can it be assumed that $K_S = K_M$ under the conditions assumed by the model?

Suppose the initial substrate concentration $[S]_0$ is assumed to be much greater than the initial enzyme concentration $[E]_0$. In that case, the formation of the enzyme-substrate complex will result in a negligible change in the free substrate concentration. Therefore, we can replace $[S] = [S]_0$ in the previous expression. Taking into account that initial velocities are often determined, equation (1) becomes:

$$v_0 = \frac{k_2[E]_0[S]_0}{K_S + [S]_0} \quad (2)$$

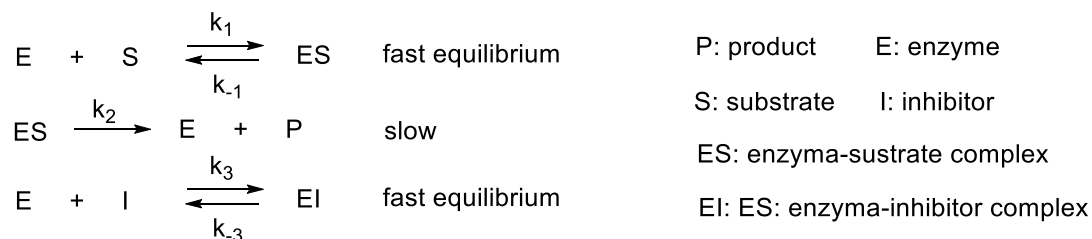
10.4. From equation (2), find an expression that describes the reaction rate v_0 i) at very low substrate concentrations and ii) at very high substrate concentrations (maximum rate, $v_{\max}$). Plot the approximate shape of a v_0 versus $[S]_0$ graph.



A competitive inhibitor is a substance that combines with the free enzyme to form a new enzyme-inhibitor EI complex and blocks the binding of the enzyme to the substrate. Under these conditions, the mass balance equation takes the form:

$$[E]_0 = [E] + [ES] + [EI]$$

and the reactions that occur are the following:



The constants K_S and K_M are defined as described above, while the new dissociation constant of the enzyme-inhibitor complex is defined as: $\frac{[E][I]}{[EI]} = \frac{k_{-3}}{k_3} = K_I$

10.5. Find an expression analogous to equation (1) for the rate law in a system where the enzyme, the substrate, and a competitive inhibitor are present. Assume the substrate concentration is much greater than the enzyme and inhibitor concentrations.

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Avogadro's constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Ideal gas equation	$p \cdot V = n \cdot R \cdot T$
Gases' constant	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	First Principle of Thermodynamic	$Q = \Delta U + W$
Zeroth in Celsius's scale	273,15 K	Work	$W = p \cdot \Delta V = F \cdot d$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Gibbs's energy	$\Delta G = \Delta H - T \cdot \Delta S$
Ionic product of water at 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \left(\frac{c_{\text{red}}}{c_{\text{ox}}} \right)$
Arrhenius' equation	$k = A \cdot \exp \left(-\frac{E_A}{R \cdot T} \right)$	$\Delta_r G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$	
First order kinetics	$\ln c_t = \ln c_0 - k_r \cdot t$	Van't Hoff factor	$i = 1 + \alpha \cdot (v - 1)$
	$t_{1/2} = \frac{\ln 2}{k_r}$	Cryoscopy	$\Delta T_{\text{crio}} = i \cdot K_{\text{crio}} \cdot b$
Absorbance	$A_\lambda = -\log_{10} \left(\frac{I}{I_0} \right) = \varepsilon \cdot l \cdot c$	Vapor pressure depression	$\Delta p = \frac{i \cdot x_s}{x_{\text{dis}} + \sum i \cdot x_s} \cdot p^\circ_{\text{dis}}$
Volume of the sphere	$V_{\text{sp}} = \frac{4}{3} \cdot \pi \cdot r^3$	Standard conditions	$p^\circ = 1 \text{ bar}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$
Area of the sphere	$A_{\text{sp}} = 4 \cdot \pi \cdot r^2$	Energy of a photon	$E = h \cdot \nu = h \cdot \frac{c}{\lambda}$
Area of the circle	$A_{\text{sp}} = \pi \cdot r^2$	Light speed	$c = 2,998 \cdot 10^8 \text{ m} \cdot \text{s}^{-1}$
Length of the circumference	$l_c = 2 \cdot \pi \cdot r$	Planck's constant	$h = 6,626 \cdot 10^{-34} \text{ J} \cdot \text{s}$
Force of gravity	$F_g = m \cdot g$	Gravity acceleration on the Earth	$g = 9,81 \text{ m} \cdot \text{s}^{-2}$
$1 \text{ J} = 1 \text{ C} \cdot \text{V} = 1 \text{ A} \cdot \text{s} \cdot \text{V} = 1 \text{ kPa} \cdot \text{L} = 1 \text{ Pa} \cdot \text{m}^3 = 1 \text{ N} \cdot \text{m} = 1 \text{ W} \cdot \text{s} = 0,2389 \text{ cal}$			
$1 \text{ atm} = 760 \text{ Torr} = 760 \text{ mm} = 101,325 \text{ kPa}$ $1 \text{ bar} = 100 \text{ kPa}$ $1 \text{ eV} = 96,485 \text{ kJ} \cdot \text{mol}^{-1}$			
$\pi = 3,1416$ $1 \text{ pm} = 0,01 \text{ \AA} = 10^{-12} \text{ m}$ $1 \text{ dm}^3 = 1 \text{ L}$ $1 \text{ mL} = 1 \text{ cm}^3$ $1 \text{ M} \equiv 1 \text{ mol/L}$			

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 H 1.008																	2 He 4.003
2	3 Li 6.941	4 Be 9.012											5 B 10.811	6 C 12.011	7 N 14.007	8 O 15.999	9 F 18.998	10 Ne 20.180
3	11 Na 22.990	12 Mg 24.305											13 Al 26.982	14 Si 28.086	15 P 30.974	16 S 32.066	17 Cl 35.453	18 Ar 39.948
4	19 K 39.098	20 Ca 40.078	21 Sc 44.956	22 Ti 47.867	23 V 50.942	24 Cr 51.996	25 Mn 54.938	26 Fe 55.845	27 Co 58.933	28 Ni 58.693	29 Cu 63.546	30 Zn 65.39	31 Ga 69.723	32 Ge 72.61	33 As 74.922	34 Se 78.96	35 Br 79.904	36 Kr 83.80
5	37 Rb 85.468	38 Sr 87.62	39 Y 88.906	40 Zr 91.224	41 Nb 92.906	42 Mo 95.94	43 Tc 98.906	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.75	52 Te 127.60	53 I 126.905	54 Xe 131.29
6	55 Cs 132.91	56 Ba 137.33	57 La 138.91	* 72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.2	83 Bi 208.98	84 Po [209]	85 At [210]	86 Rn [222]
7	87 Fr [223]	88 Ra [226]	89 Ac [227]	** 104 Rf [265]	105 Db [268]	106 Sg [271]	107 Bh [270]	108 Hs [277]	109 Mt [276]	110 Ds [281]	111 Rg [280]	112 Cn [285]	113 Nh [284]	114 Fl [289]	115 Mc [288]	116 Lv [293]	117 Ts [294]	118 Og [294]

*	58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm [145]	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.04	71 Lu 174.97
**	90 Th 232.038	91 Pa 231.036	92 U 238.029	93 Np [237]	94 Pu [242]	95 Am [243]	96 Cm [247]	97 Bk [247]	98 Cf [251]	99 Es [252]	100 Fm [257]	101 Md [258]	102 No [259]	103 Lr [262]